have to provide premiums above the current price for buy orders or discount their price for sell orders to attract additional counterparties to complete the transaction. The information content of the trade consists of inadvertently providing the market with signals to indicate the investor's buy/sell intentions, which in turn the market often interprets the stock as under-or overvalued, respectively.

Mathematically, market impact is the difference between the price trajectory of the stock with the order and what the price trajectory would have been had the order not been released to the market. Unfortunately, we are not able to simultaneously observe both price trajectories and measure market impact with any exactness. As a result, market impact has been described as the "Heisenberg uncertainty principle of trading." This concept is further described and illustrated in Chapter 4, Market Impact Models.

9. Timing Risk

Timing risk refers to the uncertainty surrounding the estimated transaction cost. It consists of three components: price volatility, liquidity risk, and parameter estimation error. Price volatility causes the underlying stock price to be either higher or lower than estimated due to market movement and noise. Liquidity risk drives market impact cost due to fluctuations in the number of counterparties in the market. Liquidity risk is dependent upon volumes, intraday trading patterns, as well as the aggregate buying and selling pressure of all market participants. Estimation error is the standard error (uncertainty) surrounding the market impact parameters.

10. Opportunity Cost

Opportunity cost is a measure of the forgone profit or avoided loss of not being able to transact the entire order (e.g., having unexecuted shares at the end of the trading period). The main reasons that opportunity cost may occur are adverse price movement and insufficient liquidity. First, if managers buy stocks that are rising, they may cancel the unexecuted shares of the order as the price becomes too expensive, resulting in a missed profit. Second, if managers cannot complete the order due to insufficient market liquidity (e.g., lack of counterparty participation) the manager would again miss out on a profit opportunity for those unexecuted shares due to favorable price movement.

TRANSACTION COST CLASSIFICATION

Transaction costs can be classified into investment related, trading related, and opportunity cost components shown above.